Griffith UNIVERSITY

3031ICT: Applied Data Mining
3804ICT: Data Mining

Lecture 6: Machine Learning in Data Mining



Dr Can Wang

School of Information and Communication Technology

Machine Learning in Data Mining

Griffith UNIVERSITY

- Machine Learning Basics
  - Basic Concepts
  - Evaluation, Overfitting & underfitting, Cross Validation
- Supervised Machine Learning Methods
  - Decision Tree Induction
  - Bayes Classification Methods
- Un-Supervised Machine Learning Methods
  - Clustering Basics
  - Partition Clustering: K-Means Clustering

Supervised vs. Unsupervised Learning

Griffith UNIVERSITY

- Supervised learning (example: classification)
  - Supervision: The training data (observations, measurements, etc.) are accompanied by **labels** indicating the class of the observations
  - New data is classified based on the training set

- Unsupervised learning (example: clustering)
  - The class labels of training data is unknown
  - Given a set of measurements, observations, etc., with the aim of establishing the existence of classes or clusters in the data

Prediction Problems: Classification vs. Numeric Prediction

Griffith UNIVERSITY

- Classification
  - predicts categorical class labels (discrete or nominal)
  - classifies data (constructs a model) based on the training set and the values (class labels) in a classifying attribute, and uses it in classifying new data
- Numerical Prediction
  - models continuous-valued functions, i.e., predicts unknown or missing values
- Typical applications
  - Credit/loan approval:
  - Medical diagnosis: if a tumor is cancerous or benign
  - Fraud detection: if a transaction is fraudulent
  - Web page categorization: which category it is

Classification—A Two-Step Process

Griffith UNIVERSITY

- Model construction: describing a set of predetermined classes
  - Each tuple/sample is assumed to belong to a predefined class, as determined by the class label attribute (decision attribute)
  - The set of tuples used for model construction is training set
- Model usage: for classifying future or unknown objects
  - Estimate accuracy of the model
    - The known label of test sample is compared with the classified result from the model
    - Accuracy rate is the percentage of test set samples that are correctly classified
    - Test set is independent of training set (otherwise overfitting)
  - If the accuracy is acceptable, use the model to classify new data

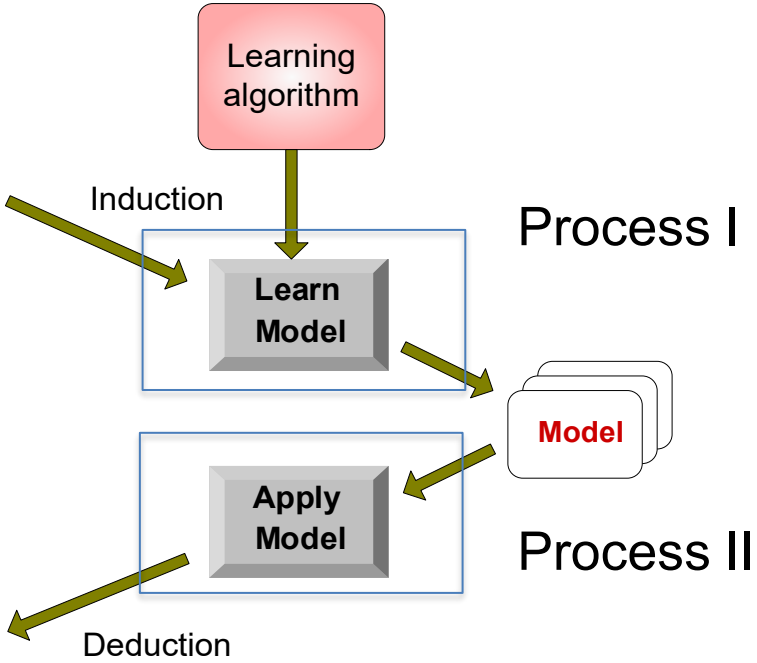
General Approach for Building Classification Model

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Evaluating Classification Methods

- **Predictive accuracy:**

$$Accuracy = \frac{\text{Number of correct classifications}}{\text{Total number of test cases}}$$

- Efficiency:
  - Time to construct the model
  - Time to use the model
- Robustness: the ability to handle noise and missing values
- Scalability: efficiency when data size increases
- Interpretability:
  - understandability and insight provided by the model
- Compactness of the model: size of the tree, or the number of rules.

Evaluation Methods

- The available dataset *D* is divided into two disjoint subsets
  - the *training set* D_{train} (for learning a model)
  - the *test set* D_{test} (for testing the model)

- **Important:** training set should not be used in testing and the test set should not be used in learning.
  - Unseen test set provides an unbiased estimate of accuracy

- The test set is also called the holdout set (If the examples in the original dataset *D* are all labeled with classes)

Precision and Recall Measures

- Used in information retrieval and text classification.
- We use a confusion matrix to introduce them.

	Classified Positive	Classified Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

where

TP: the number of correct classifications of the positive examples (**true positive**),

FN: the number of incorrect classifications of positive examples (**false negative**),

FP: the number of incorrect classifications of negative examples (**false positive**), and

TN: the number of correct classifications of negative examples (**true negative**).

Precision and Recall Measures (cont…)

	Classified Positive	Classified Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

$$p = \frac{TP}{TP + FP}. \qquad r = \frac{TP}{TP + FN}.$$

- Precision *p* is the number of correctly classified positive examples divided by the total number of examples that are classified as positive.
- Recall *r* is the number of correctly classified positive examples divided by the total number of actual positive examples in the test set.

F_1-Score

- It is hard to compare two classifiers using two measures. F_1 score combines precision and recall into one measure

$$F_1 = \frac{2pr}{p+r}$$

F_1-score is the harmonic mean of precision and recall.

$$F_1 = \frac{2}{\frac{1}{p} + \frac{1}{r}}$$

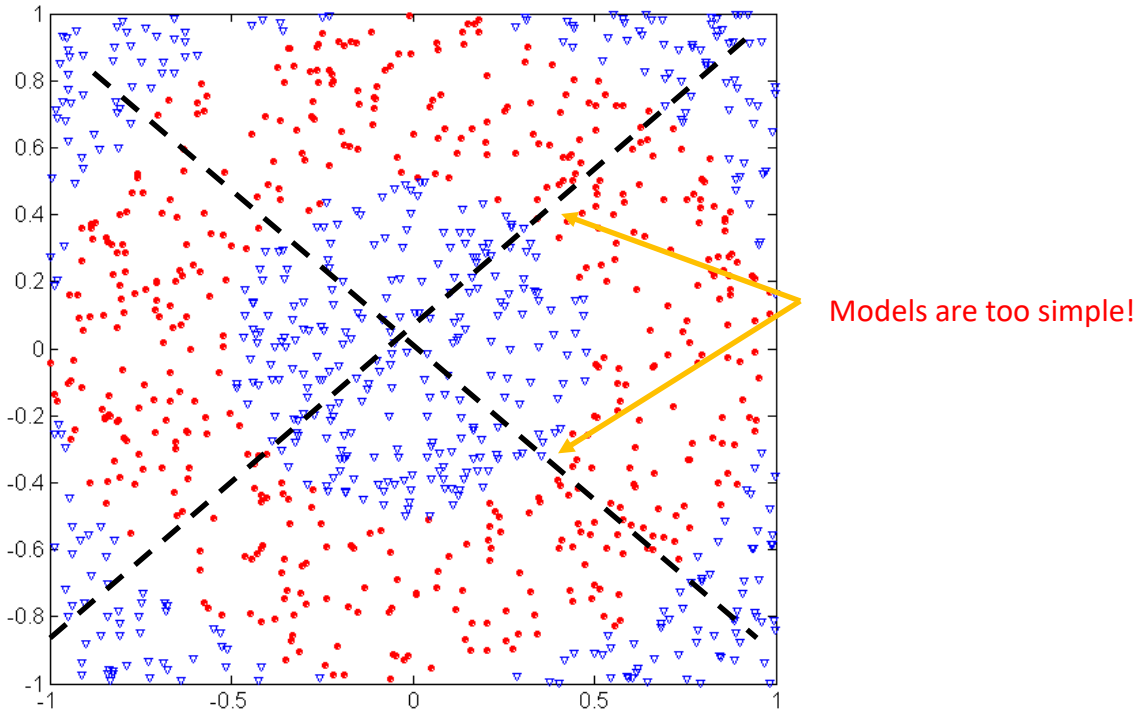
- The harmonic mean of two numbers tends to be closer to the smaller of the two.
- If F_1-value is large, both *p* and *r* must be large.

Underfitting and Overfitting

- Training errors: the number of classification errors on training records.
- Generalization (test) errors: the number of classification errors on test records.
- A good model should have low errors of both types.
- Model underfitting: model is too simple such that both Training error and Generalization error are high.
- Model overfitting: A model that fits the training data too well (with too low training errors) but may have a poorer generalization error than a model with higher training error.

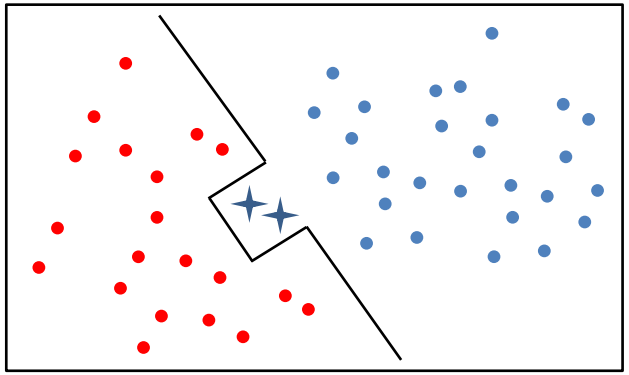
Underfitting

- Too high training errors



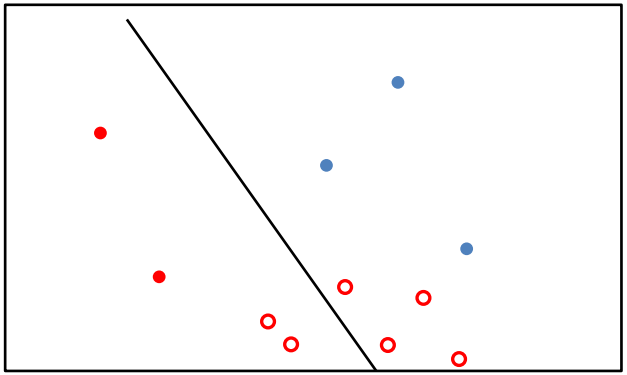
Overfitting

- Overfitting due to noise



Decision boundary is distorted by noise point

- Overfitting due to insufficient data



Lack of data points makes it difficult to draw the decision boundary (*border line between two neighboring regions of different classes*)

Evaluation methods (cont…)

- **n-fold cross-validation**: The available data is partitioned into *n* equal-size disjoint subsets.
- Use each subset as the test set and combine the rest *n*-1 subsets as the training set to learn a classifier.
- The procedure runs *n* times, which give *n* accuracies.
- The final estimated accuracy of learning is the average of the *n* accuracies.
- 10-fold and 5-fold cross-validations are commonly used (take turns to be test set).This method is used when the available data is large.

Evaluation methods (cont…)

- **Leave-one-out cross-validation**: This method is used when the data set is very small.
- It is a special case of n-fold cross-validation (Here, n is the number of data objects)
- Each fold of the cross validation has only a single test example and all the rest of the data is used in training.
- If the original data has *n* examples, this is *n*-fold cross-validation

Machine Learning in Data Mining

Griffith UNIVERSITY

- Machine Learning Basics
  - Basic Concepts
  - Evaluation, Overfitting & underfitting, Cross Validation
- Supervised Machine Learning Methods
  - Decision Tree Induction
  - Bayes Classification Methods
- Un-Supervised Machine Learning Methods
  - Clustering Basics
  - Partition Clustering: K-Means Clustering

17

Decision Tree

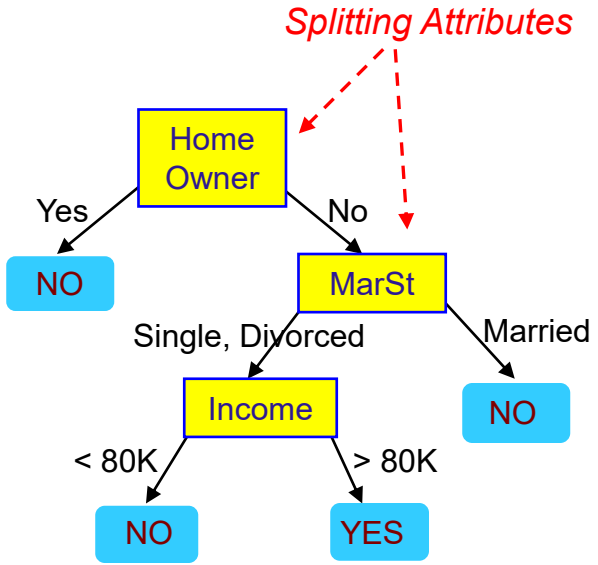
- Decision tree learning is one of the most widely used techniques for classification.
  - Its classification accuracy is competitive with other methods, and
  - It is very efficient
- The classification model is a tree, called decision tree.
- C4.5 by Ross Quinlan is perhaps the best known system. It can be downloaded from the Web.

18

Example of a Decision Tree

	categorical	categorical	continuous	class
ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

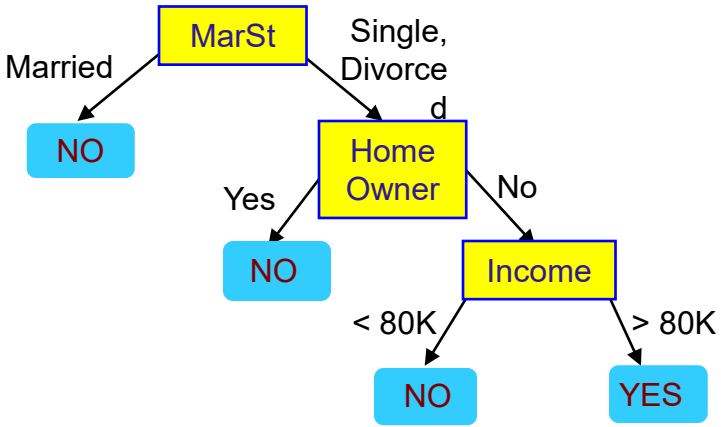
Training Data



Model: Decision Tree

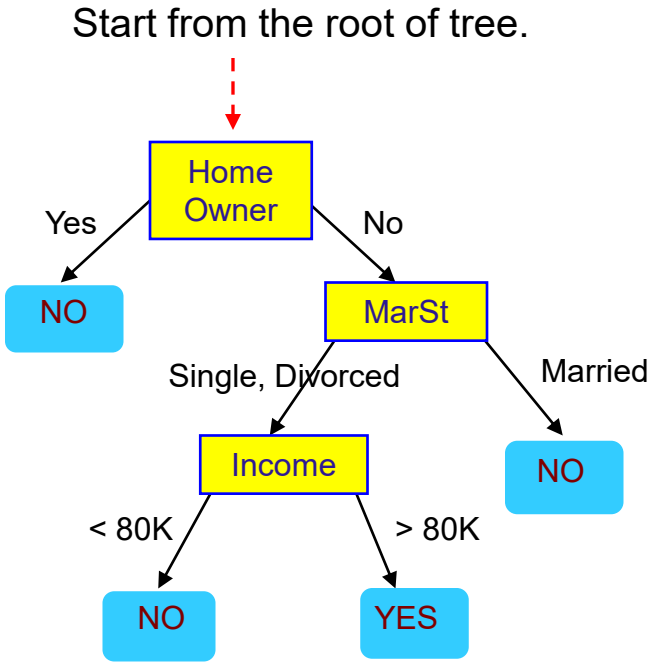
Another Example of Decision Tree

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
	categorical	categorical	continuous	class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



There could be more than one tree that fits the same data!

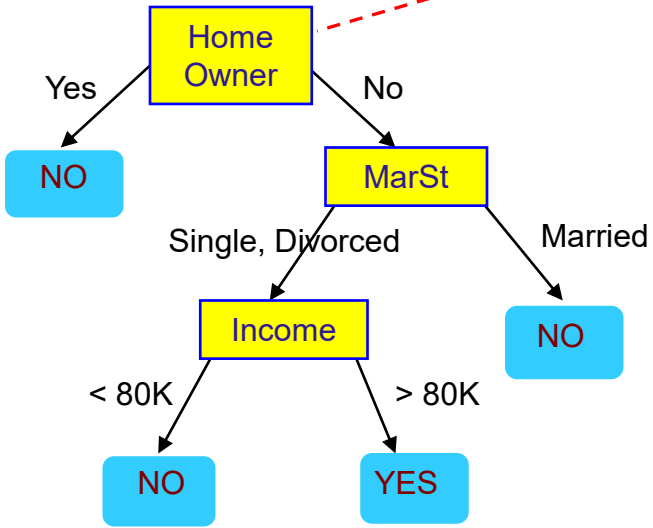
Apply Model to Test Data



Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

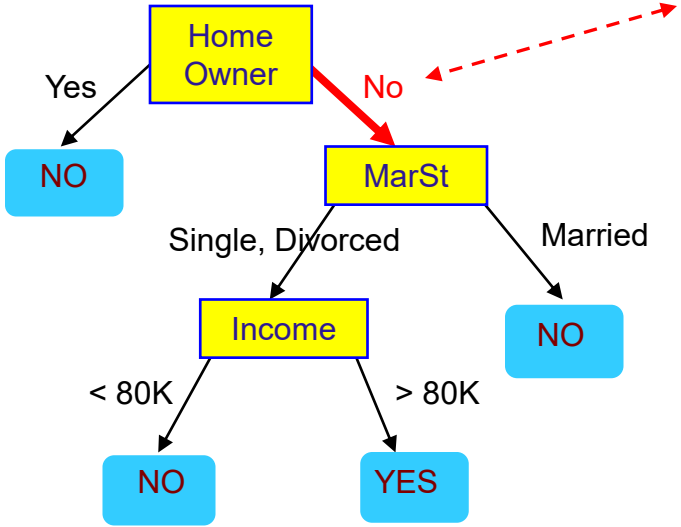
Apply Model to Test Data



Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

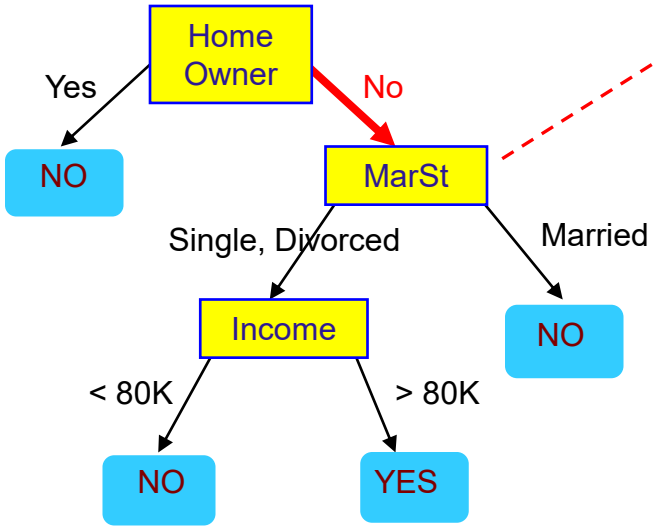
Apply Model to Test Data



Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

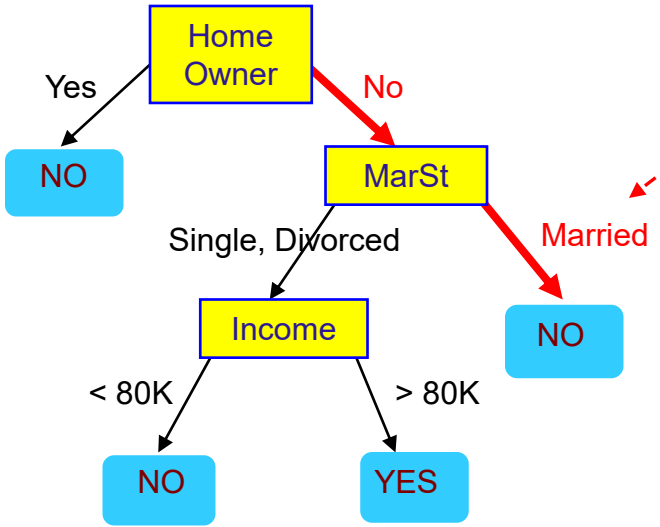
Apply Model to Test Data



Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

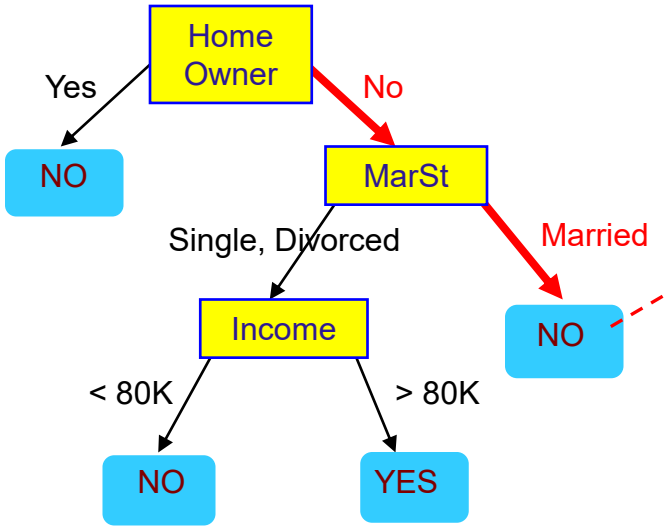
Apply Model to Test Data



Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

Apply Model to Test Data



Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

Assign Defaulted to "No"

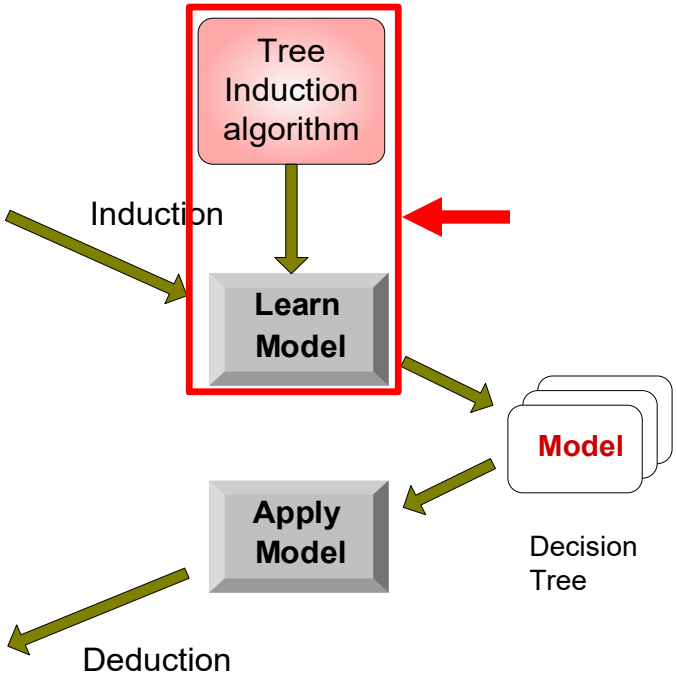
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Machine Learning in Data Mining

Griffith UNIVERSITY

- Machine Learning Basics
  - Basic Concepts
  - Evaluation, Overfitting & underfitting, Cross Validation
- Supervised Machine Learning Methods
  - Decision Tree Induction
  - Bayes Classification Methods
- Un-Supervised Machine Learning Methods
  - Clustering Basics
  - Partition Clustering: K-Means Clustering

28

Bayesian Classification: Why?

- A statistical classifier: performs *probabilistic prediction, i.e.,* predicts class membership probabilities
- Foundation: Based on Bayes' Theorem.
- Performance: A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with decision tree and selected neural network classifiers
- Incremental: Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data
- Standard: Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

Bayes Classifier

A probabilistic framework for solving classification problems

Conditional Probability:

$$P(Y \mid X) = \frac{P(X,Y)}{P(X)}$$

$$P(X \mid Y) = \frac{P(X,Y)}{P(Y)}$$

Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

Bayes Classifier

- Given new value of attributes $a_1,..., a_n$, estimate the probabilities of the class attribute taking each possible value. Choose the value with the maximum probability

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

Estimate conditional probabilities

P1= P(cheat=yes| Ref=No, MS=Married, Income=80)

P2= P(cheat=No| Ref=No, MS=Married, Income=80)

How to estimate the probabilities?

If P1>P2, classify as cheat, otherwise no cheat

Bayes Classifier

- Given feature vector A=(a_1,..., a_n), how to estimate the probability P(c|A)?
  - E.g., how to estimate P(cheat=yes| Refund=No, MStatus=Married, Income=80) or shorthand: P(yes | (No, Married, 80)) ?
- Use the Bayes theorem

$$P(c|A) = \frac{P(A|c)P(c)}{P(A)}$$

- P(c) can be estimated as the fraction of training records taking class value c, i.e., P(c)=N_c/N, where N is the total number of training records, N_c is the number of records in class c.
- No need to estimate P(A) for comparisons of P(c_1|A) and P(c_2|A)
- Thus, the problem is "how to estimate P(A|c_k)P(c_k)"?

Bayes Classifier

- Assume the features $a_1,\ldots, a_n$ are independent, given c classes, thus
- $P(C_k)P(A|C_k) = \mathrm{P}(\mathrm{A}, C_k) = \mathrm{P}(a_1, \ldots, a_n, C_k)$ ← Use Chain Rule, then we have

$$= P(a_1|a_2, \ldots, a_n, C_k) \times P(a_2, \ldots, a_n, C_k)$$

$$= P(a_1|a_2, \ldots, a_n, C_k) \times P(a_2|a_3, \ldots, a_n, C_k)P(a_3, \ldots, a_n, C_k)$$

$$= \cdots$$

$$= P(a_1|a_2, \ldots, a_n, C_k) \times P(a_2|a_3, \ldots, a_n, C_k) \times \cdots \times P(a_{n-1}|a_n, C_k) \times P(a_n|C_k)$$

$$\times P(C_k)$$

Independent Features

$$= P(a_1|C_k) \times P(a_2|C_k) \times \cdots \times P(a_n|C_k) \times P(C_k) = P(C_k) \prod_{i=1}^{n} P(a_i|C_k)$$

$\frac{N_k}{N}$ $\frac{N_{ik}}{N_k}$

N: Total number of data

N_k: Total number of k-th class data

N_{ik}: Total number of i-th feature in the k-th class data

Bayes Classifier

- Assume the features $a_1,\ldots, a_n$ are independent given class c. Thus

$$
\begin{aligned}
& P(C_k)P(A|C_k) \\
&= \mathrm{P}(\mathrm{A}, C_k) \\
&= \mathrm{P}(a_1, \ldots, a_n, C_k) \\
&= P(a_1|a_2, \ldots, a_n, C_k) \times P(a_2, \ldots, a_n, C_k) \\
&= P(a_1|a_2, \ldots, a_n, C_k) \times P(a_2|a_3, \ldots, a_n, C_k)P(a_3, \ldots, a_n, C_k) \\
&= \cdots \\
&= P(a_1|a_2, \ldots, a_n, C_k) \times P(a_2|a_3, \ldots, a_n, C_k) \times \cdots \times P(a_{n-1}|a_n, C_k) \times P(a_n|C_k) \times P(C_k) \\
&= P(a_1|C_k) \times P(a_2|C_k) \times \cdots \times P(a_n|C_k) \times P(C_k) \\
&= P(C_k) \prod_{i=1}^{n} P(a_i|C_k)
\end{aligned}
$$

Use Chain Rule, then we have

Independent Features

$\frac{N_k}{N}$ $\frac{N_{ik}}{N_k}$

N: Total number of data

N_k: Total number of k-th class data

N_{ik}: Total number of i-th feature in the k-th class data

Bayes Classifier Example

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

Test data: A: (yes, no, yes, no)

M: mammals

N: non-mammals

$$P(A|M)P(M) \; vs \; P(A|N)P(N)$$

$$P(A|M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A|N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A|M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A|N)P(N) = 0.0042 \times \frac{13}{20} = 0.0027$$

$$P(A|M)P(M) > P(A|N)P(N)$$

=> Mammals

Machine Learning in Data Mining

- Machine Learning Basics
  - Basic Concepts
  - Evaluation, Overfitting & underfitting, Cross Validation
- Supervised Machine Learning Methods
  - Decision Tree Induction
  - Bayes Classification Methods
- Un-Supervised Machine Learning Methods
  - Clustering Basics
  - Partition Clustering: K-Means Clustering

36

What is Cluster Analysis?

- Cluster: A collection of data objects
  - similar (or related) to one another within the same group
  - dissimilar (or unrelated) to the objects in other groups
- Cluster analysis (or *clustering*, *data segmentation, …*)
  - Finding similarities between data according to the characteristics found in the data and grouping similar data objects into clusters
- Unsupervised learning: no predefined classes (i.e., *learning by observations* vs. learning by examples: supervised)
- Typical applications
  - As a stand-alone tool to get insight into data distribution
  - As a preprocessing step for other algorithms

Clustering for Data Understanding and Applications

Griffith UNIVERSITY

- Biology: taxonomy of living things -- kingdom, phylum, class, order, family, genus and species
- Information retrieval: document clustering
- Land use: Identification of areas of similar land use in an earth observation database
- Marketing: Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs
- City-planning: Identifying groups of houses according to their house type, value, and geographical location
- Earth-quake studies: Observed earthquake epicenters should be clustered along continent faults
- Climate: understanding earth climate, find patterns of atmosphere and ocean
- Economic Science: market research

Quality: What Is Good Clustering?

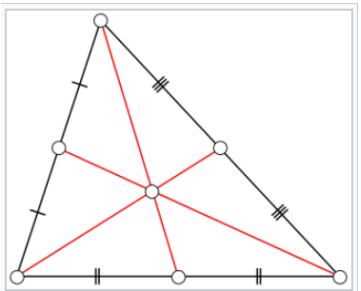
Griffith UNIVERSITY

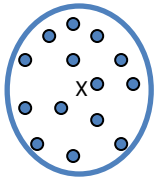
- A good clustering method will produce high quality clusters
  - high intra-class similarity: cohesive within clusters
  - low inter-class similarity: distinctive between clusters

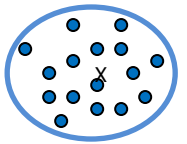
- The quality of a clustering method depends on
  - the similarity measure used by the method
  - its implementation, and
  - Its ability to discover some or all of the hidden patterns

Distance between Clusters

- **Single link:** the smallest distance between an element in one cluster and an element in the other, i.e., $dist(K_i, K_j) = min(t_{ip}, t_{jq})$
- **Complete link:** the largest distance between an element in one cluster and an element in the other, i.e., $dist(K_i, K_j) = max(t_{ip}, t_{jq})$
- **Average:** the avg distance between an element in one cluster and an element in the other, i.e., $dist(K_i, K_j) = avg(t_{ip}, t_{jq})$
- **Centroid:** the distance between the centroids of two clusters, i.e., $dist(K_i, K_j) = dist(C_i, C_j)$
- **Medoid:** the distance between the medoids of two clusters, i.e., $dist(K_i, K_j) = dist(M_i, M_j)$
  - Medoid: a chosen, centrally located object in the cluster


Centroid of a triangle





40

Major Clustering Approaches (I)

- Partitioning approach:
  - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
  - Typical methods: **k-means**, k-medoids, CLARANS
- Hierarchical approach:
  - Create a hierarchical decomposition of the set of data (or objects) using some criterion
  - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Density-based approach:
  - Based on connectivity and density functions
  - Typical methods: **DBSCAN**, OPTICS, DenClue
- Grid-based approach:
  - Based on a multiple-level granularity structure
  - Typical methods: STING, WaveCluster, CLIQUE

Major Clustering Approaches (II)

- Model-based clustering:
  - A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other
  - Typical methods: EM, SOM, COBWEB
- Frequent pattern-based clustering:
  - Based on the analysis of frequent patterns
  - Typical methods: p-Cluster
- User-guided or constraint-based clustering:
  - Clustering by considering user-specified or application-specific constraints
  - Typical methods: COD (obstacles), constrained clustering
- Link-based clustering:
  - Objects are often linked together in various ways
  - Massive links can be used to cluster objects: SimRank, LinkClus

Machine Learning in Data Mining

Griffith UNIVERSITY

- Machine Learning Basics
  - Basic Concepts
  - Evaluation, Overfitting & underfitting, Cross Validation
- Supervised Machine Learning Methods
  - Decision Tree Induction
  - Bayes Classification Methods
- Un-Supervised Machine Learning Methods
  - Clustering Basics
  - Partition Clustering: K-Means Clustering

43

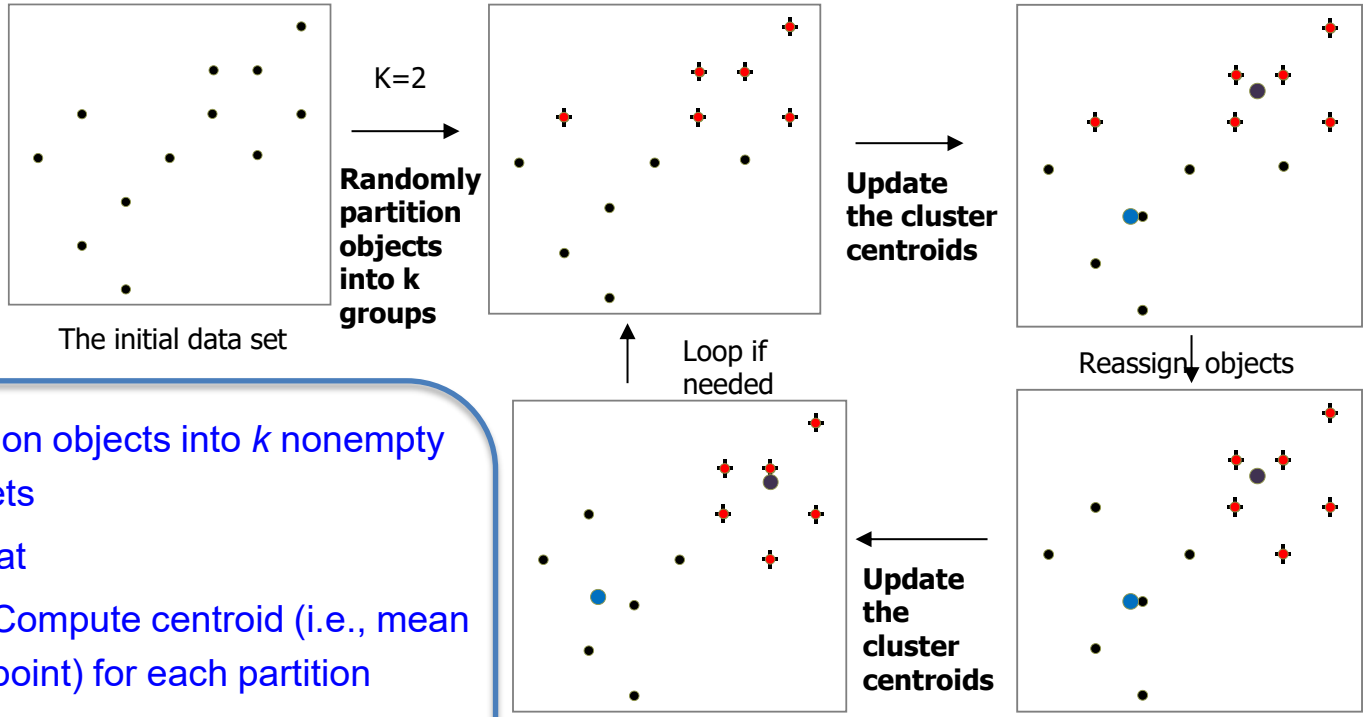
Partitioning Algorithms

- Partitioning method: Partitioning a database ***D*** of ***n*** objects into a set of ***k*** clusters, such that the sum of squared distances is minimized (where c_i is the centroid or medoid of cluster C_i)

$$E = \Sigma_{i=1}^{k} \Sigma_{p \in C_i} (p - c_i)^2$$

- Given *k*, the *k-means* algorithm is implemented in four steps:
  - Partition objects into *k* nonempty subsets
  - Compute seed points as the centroids of the clusters of the current partitioning (the centroid is the center, i.e., *mean point*, of the cluster)
  - Assign each object to the cluster with the nearest seed point
  - Go back to Step 2, stop when the assignment does not change

An Example of *K-Means* Clustering



- Partition objects into *k* nonempty subsets
- Repeat
  - Compute centroid (i.e., mean point) for each partition
  - Assign each object to the cluster of its nearest centroid
- Until no change

Machine Learning in Data Mining

Griffith UNIVERSITY

- Machine Learning Basics
  - Basic Concepts
  - Evaluation, Overfitting & underfitting, Cross Validation
- Supervised Machine Learning Methods
  - Decision Tree Induction
  - Bayes Classification Methods
- Un-Supervised Machine Learning Methods
  - Clustering Basics
  - Partition Clustering: K-Means Clustering

46

Have a question?

